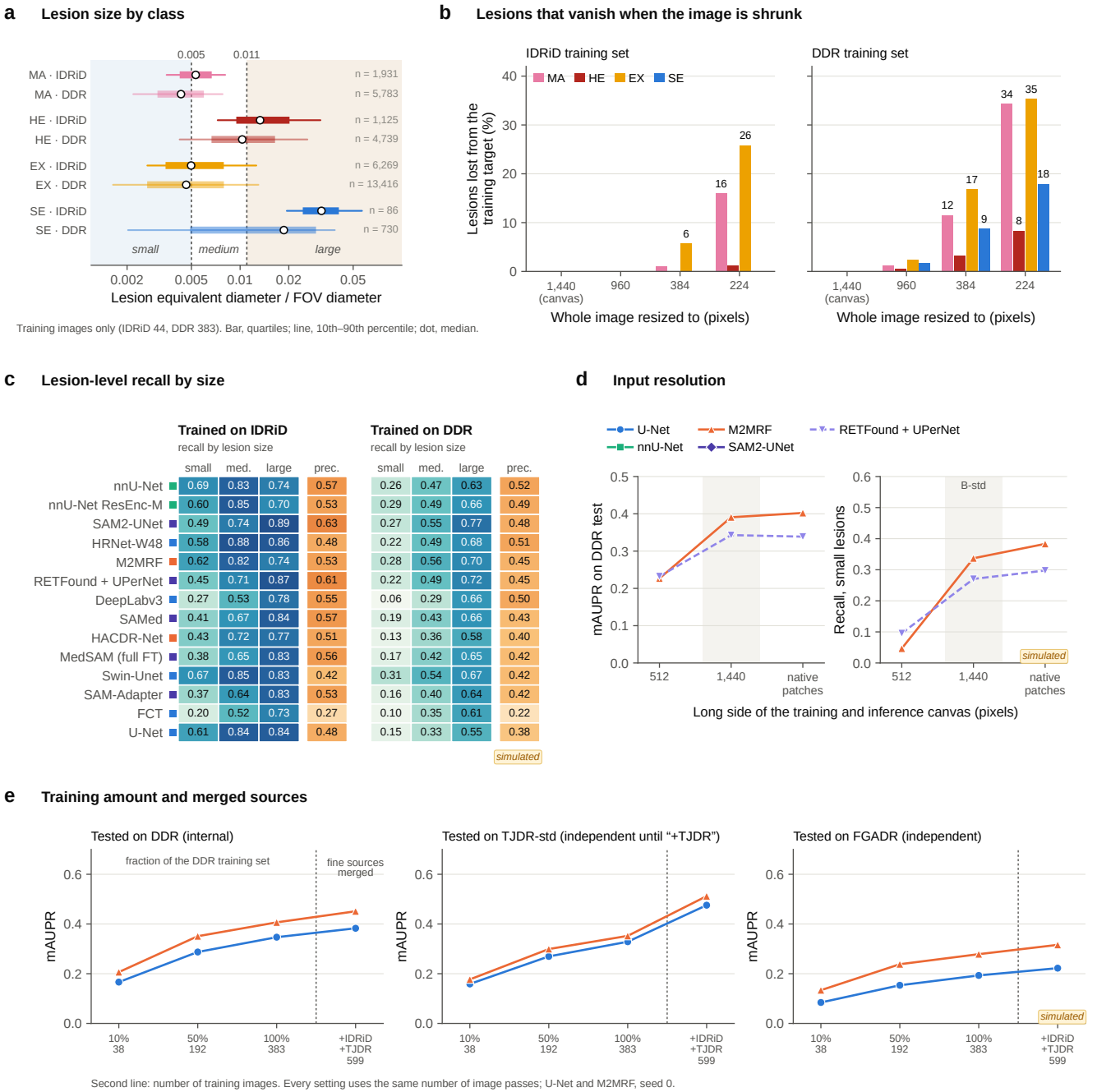
a
Lesion size by class
0.005
0.011
MA · IDRiD
MA · DDR
HE · IDRiD
HE · DDR
EX · IDRiD
EX · DDR
SE · IDRiD
SE · DDR
n = 1,931
n = 5,783
n = 1,125
n = 4,739
n = 6,269
n = 13,416
n = 86
n = 730
small
medium
large
0.002
0.005
0.01
0.02
0.05
Lesion equivalent diameter / FOV diameter
b
Lesions that vanish when the image is shrunk
IDRiD training set
DDR training set
MA
HE
EX
SE
Lesions lost from the training target (%)
0
10
20
30
40
6
16
26
12
17
9
34
8
35
18
1,440 (canvas)
960
384
224
Whole image resized to (pixels)
Training images only (IDRiD 44, DDR 383). Bar, quartiles; line, 10th–90th percentile; dot, median.
c
Lesion-level recall by size
Trained on IDRiD
recall by lesion size
Trained on DDR
recall by lesion size
small
med.
large
prec.
nnU-Net 0.69 0.83 0.74 0.57 | 0.26 0.47 0.63 0.52
nnU-Net ResEnc-M 0.60 0.85 0.70 0.53 | 0.29 0.49 0.66 0.49
SAM2-UNet 0.49 0.74 0.89 0.63 | 0.27 0.55 0.77 0.48
HRNet-W48 0.58 0.88 0.86 0.48 | 0.22 0.49 0.68 0.51
M2MRF 0.62 0.82 0.74 0.53 | 0.28 0.56 0.70 0.45
RETFound + UPerNet 0.45 0.71 0.87 0.61 | 0.22 0.49 0.72 0.45
DeepLabv3 0.27 0.53 0.78 0.55 | 0.06 0.29 0.66 0.50
SAMed 0.41 0.67 0.84 0.57 | 0.19 0.43 0.66 0.43
HACDR-Net 0.43 0.72 0.77 0.51 | 0.13 0.36 0.58 0.40
MedSAM (full FT) 0.38 0.65 0.83 0.56 | 0.17 0.42 0.65 0.42
Swin-Unet 0.67 0.85 0.83 0.42 | 0.31 0.54 0.67 0.42
SAM-Adapter 0.37 0.64 0.83 0.53 | 0.16 0.40 0.64 0.42
FCT 0.20 0.52 0.73 0.27 | 0.10 0.35 0.61 0.22
U-Net 0.61 0.84 0.84 0.48 | 0.15 0.33 0.55 0.38
simulated
d
Input resolution
U-Net
M2MRF
RETFound + UPerNet
nnU-Net
SAM2-UNet
mAUPR on DDR test
Recall, small lesions
B-std
0.0
0.1
0.2
0.3
0.4
0.5
0.6
512
1,440
native patches
simulated
Long side of the training and inference canvas (pixels)
e
Training amount and merged sources
Tested on DDR (internal)
Tested on TJDR-std (independent until “+TJDR”)
Tested on FGADR (independent)
fraction of the DDR training set
fine sources merged
mAUPR
0.0
0.2
0.4
0.6
10% 38
50% 192
100% 383
+IDRiD +TJDR 599
simulated
Second line: number of training images. Every setting uses the same number of image passes; U-Net and M2MRF, seed 0.